

Specifications

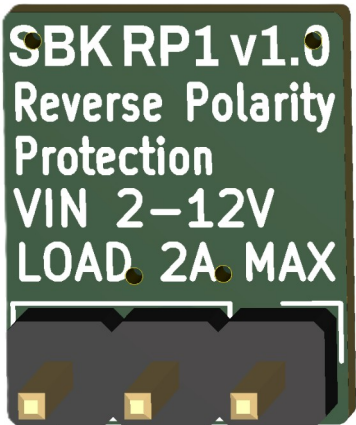
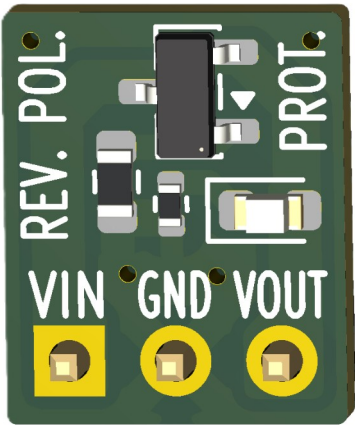
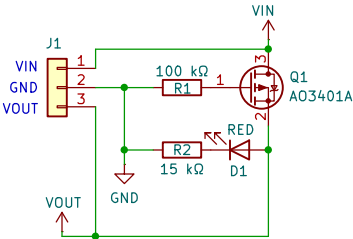
Input voltage	2-12 VDC
Maximum load current	2 A
Topology	P-channel MOSFET
Forward voltage drop	< 100 mV*
Typical quiescent current	< 1 µA
PCB dimensions	(your PCB size)
Connector pitch	2.54 mm

* Depends on load current, MOSFET characteristics and temperature.

Features

- Reverse polarity protection for low-voltage DC circuits
- P-channel MOSFET ideal-diode topology
- Very low voltage drop
- Low power dissipation
- Output power status LED
- Breadboard compatible
- Open-source hardware

BOM				
Reference	Qty	Value	Package	Description
D1	1	RED	0603	Output status LED
J1	1	1 x 3 pin header	2.54 mm	3-pin header
Q1	1	A03401A	SOT-23	P-channel MOSFET
R1	1	100 kΩ	0603	Gate protection resistor
R2	1	15 kΩ	0402	Output status LED resistor



SBK_RP1 - Reverse Polarity Protection

Licensed under CERN-OHL-P-2.0

Smart Builds & Kits

Sheet: /
File: SBK_RP1.kicad_sch

Title: **SBK_RP1 - Reverse Polarity Protection**

Size: USLetter Date: 2026-07-31
KiCad E.D.A. 10.0.3

Rev: **v1.0**
Id: 1/1